

noise at high noise prone area like compressor House in a factory.

- Q.28 Differentiate between Heat Stress heat strain.
- Q.29 Explain various types of earthing system used in electrical installations.
- Q.30 Explain the reason for use of logarithmic scale i.e. dB for noise measurement instead of an absolute scale i.e. Pa/Watt. What is difference between dB(A) & dB?
- Q.31 Define the following terms :
- a) Dust Explosion b) UEL
 - c) Flash Point
 - d) Auto Ignition Temperature
 - e) Pool Fires
- Q.32 List the measures to ensure good ventilation in an industrial building. Explain with the help of diagram exhaust ventilation system.
- Q.33 What is meant by Fire Load? Illustrate with an example the process of determining fire load of a building.
- Q.34 Explain various types of lighting with suitable examples and sketches.
- Q.35 Explain the undesirable effect of vibration on human body at workplace.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 List the various types of electrical safety devices used in electrical circuits along with their uses. Explain the working principles of ELCB and RCCB with sketches.
- Q.37 Explain in detail the phenomena of BLEVE. What are the causes and consequences of BLEVE?
- Q.38 Explain the principles of good plant Layout to ensure safety at workplace.

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1st Year / Industrial Safety Subject:- Safety Engineering - II

Time : 3Hrs.

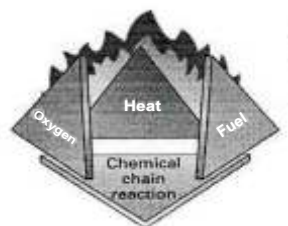
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SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 What is the full form of LOTO?
- a) Line out Tag out b) Link out Tag out
 - c) Lock out Tag out d) Limit out Tag out
- Q.2 The equipment mounted on building for lightening protection
- a) ELCB b) RCCB
 - c) Lightning Arrestor d) MCCB
- Q.3 Fall protection systems can include which three of the following:
- a) Guardrail systems
 - b) Safety nets
 - c) Off-site video monitoring
 - d) Both a & b
- Q.4 The noise level measure widely used to evaluate occupational exposure because of its good correlation with hearing damage;
- a) A-weighted level" and expressed as dB(A)
 - b) C-weighted level" and expressed as dB(C)
 - c) B-weighted level" and expressed as dB(B)
 - d) All of the above
- Q.5 Gas Test should mandatorily done before start of
- a) Confined space works

- b) Excavation work greater than 1.5 m
 c) Cold works
 d) Both (a) & (b)
- Q.6 Which of the following is Code of Practice - Work Permit System?
 a) IS 17893 b) IS 18793
 c) IS 19823 d) IS 17890
- Q.7 The maximum distance where the crane boom has to reach for lifting or lowering the load
 a) Distance Arm b) Working Arm
 c) Boom Length d) SWL
- Q.8 The name of safety device used in telescopic crane for measurement of wind speed
 a) Ammeter b) Anemometer
 c) Hygrometer d) Safe Load indicator
- Q.9 Which member of scaffold responsible for resisting lateral loads?
 a) Bracing b) Standards
 c) Lifeline d) Transom
- Q.10 Identify the name of the theory/model represented by the figure above, which explains the basic elements necessary for a fire to occur and sustain itself.



- a) Fire Quadrilateral Theory
 b) Fire Tetrahedron Theory
 c) Fire Triangle Theory
 d) Fire Pentagon Theory

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SECTION-B

Note: Objective type questions. All questions are compulsory.
 (10x1=10)

- Q.11 What do you mean by Engineering Controls in risk Management?
- Q.12 What is meant by Natural Ventilation?
- Q.13 Give full form of WBGT.
- Q.14 What is Whole Body Vibration?
- Q.15 What is meant by earthing and its significance in electrical safety.
- Q.16 Explain vibration isolation of machine foundation.
- Q.17 What are the hazards associated with borrowed neutrals?
- Q.18 Write a short note on Glare and its control measures.
- Q.19 Explain the importance of Bonding in avoiding static electricity.
- Q.20 What is meant by Noise Induced Hearing Loss?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions.
 (12x5=60)

- Q.21 Explain Risk Matrix used while carrying out HIRA and its significance.
- Q.22 Elaborate the principles of good housekeeping to ensure safety at workplace.
- Q.23 Describe the various undesirable effects of improper lighting at workplace and its control measures.
- Q.24 Elaborate the factors to be considered while preparation of Emergency Response plan for handling onsite and offsite emergencies of hazardous process industry.
- Q.25 Explain hazardous zone classification in a factory and its significance for job specific risk assessment.
- Q.26 Write a note on types of portable fire extinguisher and their applications.
- Q.27 Write a short note on preventive measures to control

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